Jeff Disbrow-Monz



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github.com/disbr007

Languages

Python	TypeScript
SQL	Bash

Databases

Postgres	Firebase
PostGIS	DynamoDB

Messaging / Events

NATS	Firebase Queue
SQS/SNS	

DevOps + Cloud

Docker	Kubernetes
EC2	RDS
EKS	S3
ECR	Cloudfront
GitHub Actions	Grafana

Software Engineer focused on building the software and data systems needed to tackle climate change. I've led and contributed across the full development lifecycle: partnering on product strategy, designing architecture, and handling engineering, deployment, and monitoring. With a geospatial background, I specialize in distilling complex environmental datasets into relevant, usable information. I am a strong advocate for engineering excellence, ensuring that as systems grow in complexity, they remain reliable, maintainable, and aligned with long-term technical goals.

EXPERIENCE

Principal Software Engineer, PVcase
Mar 2025 – Present

- Led the creation, review and approval of architectural decisions via ADRs, ensuring alignment between product requirements and technical constraints.
- Developed the strategy and tooling to perform application data migrations.
- Refactored a legacy task processing service, removing a deprecated core dependency and transitioning to worker processes pulling from a NATS queue, ultimately improving task processing speed.
- Partnered with engineering leadership to define product-wide KPIs and SLOs, using a data-driving framework to monitor and improve system reliability and reduce critical incidents.

Software Engineer | Engineering Manager, PVcase
Jan 2021 – Mar 2025

- Designed and built an ETL pipeline and backend service to ingest, normalize, and expose third-party RTO/ISO capacity data.
- Developed a GIS data ingestion service which allowed users to drag-and-drop their data, reducing manual data processing time by 70%.
- Spearheaded technical product expansion into Europe under an aggressive timeline, adapting legacy systems to support international data sources.
- Led a full-stack engineering and QA team, overseeing the end-to-end life cycle, deployment strategy, and monitoring of critical services.
- Collaborated with Product leadership to prioritize tasks and refine the team's work stream.

Graduate Research Assistant, Polar Geospatial Center
Oct 2018 – Jan 2021

- Built a pipeline for selecting, ordering, downloading, and archiving satellite imagery to create stereoscopic digital elevation models.
- Developed an algorithm to select imagery for Arctic coastline mapping by merging imagery metadata and external sea-ice concentration.
- Created derivative products from raw satellite imagery, including orthorectified and pansharpened images, digital elevation models, and cartographic products.

GIS Analyst / Cartographer, Northeastern Minnesotans for Wilderness
Sept 2017 – Oct 2018

- Built a web map using ArcGIS API for JavaScript and Collector to visualize field data, including datasets from state and federal agencies, private partners, and public sources.

EDUCATION

M.S. Natural Resources Science, University of Minnesota
Dec 2023
Assessment, Monitoring, and Geospatial Analysis Track
Thesis: Retrogressive Thaw Slump Classification Using High Resolution Satellite Imagery and Stereo-Derived Digital Elevation Models

B.S. Parks, Recreation, and Tourism University of Vermont
Dec 2012